

Honors Geometry - Chapter 2 Test PRACTICE

Name: _____ Period: _____

Reasoning and Proofs • Sections 2.1 - 2.6 • same shape as the test, different numbers

Show all your work. Check your work. Write neatly. Circle your final answer.

1. Use the conditional statement: If a figure is a square, then it has four right angles. (2 pts)

a) Hypothesis: _____

Conclusion: _____

b) Write the converse. Is it true or false? If false, give a counterexample.

2. Write the inverse and the contrapositive: If it is Saturday, then there is no school. (2 pts)

Inverse:

Contrapositive:

3. The conditional statement below and its converse are both true. Write it as a biconditional statement. (1 pt)
- If two angles are complementary, then the sum of their measures is 90° .

4. Answer both parts. (2 pts)

a) Find the next item in the pattern: 7, 14, 28, 56, ...

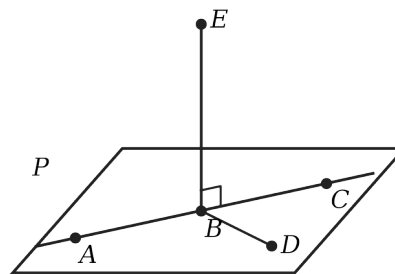
b) Find a counterexample: Every prime number is odd.

5. Is each conjecture valid? If valid, name the law. If not, explain why. (2 pts)

a) If a triangle is equilateral, then it is isosceles. Triangle JKL is equilateral. Conjecture: Triangle JKL is isosceles.

b) If an animal is a dog, then it is a mammal. If an animal is a mammal, then it has a backbone. Conjecture: If an animal has a backbone, then it is a dog.

Use the diagram for question 6.



6. Is each statement valid based on the diagram? Circle VALID or NOT VALID and explain. (2 pts)

a) Points B, D, and E are collinear.
VALID / NOT VALID because

b) Line AC lies in plane P.
VALID / NOT VALID because

7. Name the postulate. (2 pts)

a) A plane contains at least three noncollinear points.

b) Through any three noncollinear points, there exists exactly one plane.

8. Solve $4(x - 1) + 3x = 31$. Write every step and justify each one. (3 pts)

Statements	Reasons
1. $4(x - 1) + 3x = 31$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.

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Name: _____

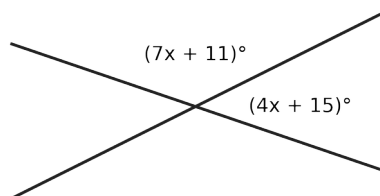
9. Name the property that justifies each statement. (2 pts)
- a) If $GH = 12$ and $12 = JK$, then $GH = JK$.

- b) If $\angle 1 \cong \angle 2$, then $\angle 2 \cong \angle 1$.

10. Ray BD bisects $\angle ABC$. $m\angle ABD = (4x + 9)^\circ$ and $m\angle DBC = (6x - 5)^\circ$. Find $m\angle ABC$. (2 pts)

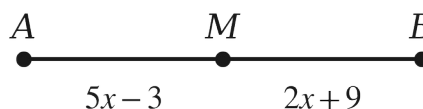
$m\angle ABC = \underline{\hspace{2cm}}^\circ$

13. Find the value of x . Then name the postulate or theorem that lets you write your equation. (2 pts)



$x = \underline{\hspace{2cm}}$ Reason: _____

11. Complete the proof. The statements are done; supply each reason. (Prove: $x = 4$) (3 pts)



Given: M is the midpoint of segment AB ;
 $AM = 5x - 3$ and $MB = 2x + 9$.
 Prove: $x = 4$

Statements	Reasons
1. M is the midpoint of AB ; $AM = 5x - 3$, $MB = 2x + 9$	1. Given
2. $AM = MB$	2.
3. $5x - 3 = 2x + 9$	3.
4. $3x - 3 = 9$	4.
5. $3x = 12$	5.
6. $x = 4$	6.

12. Write the proof. Line 1 is done. Every line needs a statement AND a reason. (5 pts)

Given: $\angle 1 \cong \angle 2$; $\angle 1$ and $\angle 3$ are supplementary.
 Prove: $\angle 2$ and $\angle 3$ are supplementary



Statements	Reasons
1. $\angle 1 \cong \angle 2$; $\angle 1$ and $\angle 3$ are supplementary	1. Given
2.	2.
3.	3.
4.	4.
5.	5.

REASON BANK — use a reason as often as you need; not every reason is used.

Given · Definition of midpoint · Definition of a segment bisector · Definition of angle bisector · Definition of congruent angles · Definition of supplementary angles · Definition of complementary angles · Definition of right angle · Definition of perpendicular · Angle Addition Postulate · Linear Pair Postulate · Vertical Angles Theorem · Congruent Supplements Theorem · Distributive Property · Simplify (combine like terms) · Addition / Subtraction / Multiplication / Division Property of Equality · Substitution Property of Equality · Reflexive Property of Equality · Symmetric Property of Equality · Transitive Property of Equality